

AUTOMATED FUNCTIONALITY DETECTION SYSTEM FOR OUTDOOR ADVERTISING

Report of Industrial Training

by

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AUTOMATED FUNCTIONALITY DETECTION SYSTEM FOR OUTDOOR ADVERTISING

*This report is submitted in a partial fulfilment of the requirements for the four years
Bachelor of Science (Joint Major) Degree
In Electronics and Computing and Information Systems*

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DECLARATION

I declare that this report, which I submit to the Wayamba University of Sri Lanka for examination in consideration of the award of the BSc. (Joint Major) Degree, is my own personal effort. Where any of the content presented is the result of input or data from a related collaborative research program this is duly acknowledged in the text such that it is possible to ascertain how much of the work is my own. Furthermore, I took reasonable care to ensure that the work is original, and to the best of my knowledge, does not breach copyright law, and has not been taken from other sources except where such work has been cited and acknowledged within the text.

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ABSTRACT

This project was undertaken at Synergy Tech Consultancy (PVT) Ltd., Nugegoda, in collaboration with the Department of Electronics, Faculty of Applied Sciences, Wayamba University of Sri Lanka, focusing on the design and implementation of an automated functionality detection system for outdoor advertising infrastructure. The primary objective was to develop a comprehensive IoT-based monitoring solution that addresses critical gaps in current outdoor advertising monitoring capabilities, specifically the lack of real-time fault detection and granular diagnostic information. The assigned responsibilities included complete system design from hardware circuit development through cloud infrastructure implementation to web application creation. Hardware development involved designing tri-level fault detection circuits using PC817 optocouplers, current transformers, and ESP32 WROOM-32 microcontroller integration with supporting power management systems. The circuit design utilized three 75k Ω resistors for voltage division, MOV surge protection, and 4007 diode bridge rectification for reliable AC power monitoring, while current detection employed 5A/5mA current transformers with burden resistors and Zener diode protection. PCB design and fabrication were completed using Altium Designer software, followed by 3D printed PLA+ enclosure design using SolidWorks for weatherproof outdoor deployment. Software development responsibilities encompassed ESP32 firmware programming using ESP-IDF framework with FreeRTOS multitasking implementation, MQTT communication protocol integration, and comprehensive web application development using React.js frontend with Firebase authentication and Node.js backend services. Cloud infrastructure implementation included Telegraf data processing agents and InfluxDB time-series database management. The system achieved exceptional performance results with 100% accuracy in fault classification testing, distinguishing successfully between commercial power failures, DC converter malfunctions, and output switching circuit problems. Communication reliability exceeded 99% data transmission success rates with average latencies of 35 seconds. Field deployment demonstrated reliable operation with 6.5-hour average battery backup capability and system uptime exceeding 99.7%. The web application provided intuitive geographical visualization through Leaflet mapping integration with real-time status updates and automated report generation in PDF and CSV formats. The learning experience gained included advanced embedded systems design, IoT communication protocols, cloud infrastructure architecture, modern web development frameworks, and professional PCB design methodologies, providing comprehensive exposure to industrial IoT system development and deployment practices.

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LIST OF ABBREVIATION

VCL: Visual Concept Labs (Pvt) Ltd.
STC: Synergy Tech Consultancy (Pvt) Ltd.
AI: Artificial Intelligence
SWOT: Strengths, Weaknesses, Opportunities, and Threats
LED: Light-Emitting Diode
IoT: Internet of Things
PDF: Portable Document Format
CSV: Comma-Separated Values
AC: Alternating Current
DC: Direct Current
MQTT: Message Queuing Telemetry Transport
JSON: JavaScript Object Notation
ID: Identifier
API: Application Programming Interface
MOV: Metal Oxide Varistor
CT: Current Transformer
SRAM: Static Random-Access Memory
ADC: Analog-to-Digital Converter
GPIO: General-Purpose Input/Output
VAC: Volts Alternating Current
VDC: Volts Direct Current
PCB: Printed Circuit Board
PLA: Polylactic Acid
UV: Ultraviolet
FreeRTOS: Free Real-Time Operating System
HTTP: Hypertext Transfer Protocol
DOM: Document Object Model
UI: User Interface
CSS: Cascading Style Sheets

CHAPTER 1: INTRODUCTION

1.1 Synergy Tech Consultancy (PVT) Ltd.

1.1.1 Background of the Company



(a)



(b)



(c)

Fig. 1.1: (a) Logo STC. (b) Logo VCL. (c) Logo Eco Pulse

Visual Concept Labs (Pvt) Ltd. (VCL) is a research-focused technology company specializing in computer vision, machine learning, artificial intelligence, and advanced computer vision-based solutions. Established with a strong emphasis on innovation, VCL operates as a research-based software developer, system integrator, and consultancy service provider in the digital media and intelligent systems domains. Headquartered in Colombo, Sri Lanka, VCL also maintains operations in the USA, Canada, and Australia, positioning itself as a global high-tech player.

Synergy Tech Consultancy (Pvt) Ltd. (STC), a subsidiary of Visual Concept Labs, serves as the bridge between cutting-edge research and real-world applications. STC primarily focuses on computer vision and machine learning, transforming VCL's research outputs into commercially viable products and solutions. Leveraging expertise in product development and market deployment, STC ensures the effective delivery of VCL's innovations to industry stakeholders.

To diversify its offerings and address emerging needs in the industry 4.0 era, Eco Pulse (Pvt) Ltd. was established under the broader vision of the group. Eco Pulse is dedicated to research and the development of innovative, research-based solutions designed to tackle complex Industry 4.0 challenges. While VCL concentrates on foundational research in AI and computer vision, and STC handles productization and commercialization, Eco Pulse functions as the specialized arm focused on designing and developing solutions tailored to Industry 4.0 applications.

Together, these three companies form a collaborative ecosystem where pioneering research, product development, and market-driven innovation work in synergy to create impactful solutions across diverse industries.

1.1.2 Company Structure

The following is a simplified hierarchy of the company. Senior management operates remotely, while most other employees work in a hybrid mode (both on-site and remote). Interns and junior developers, including new employees, work physically in the office.

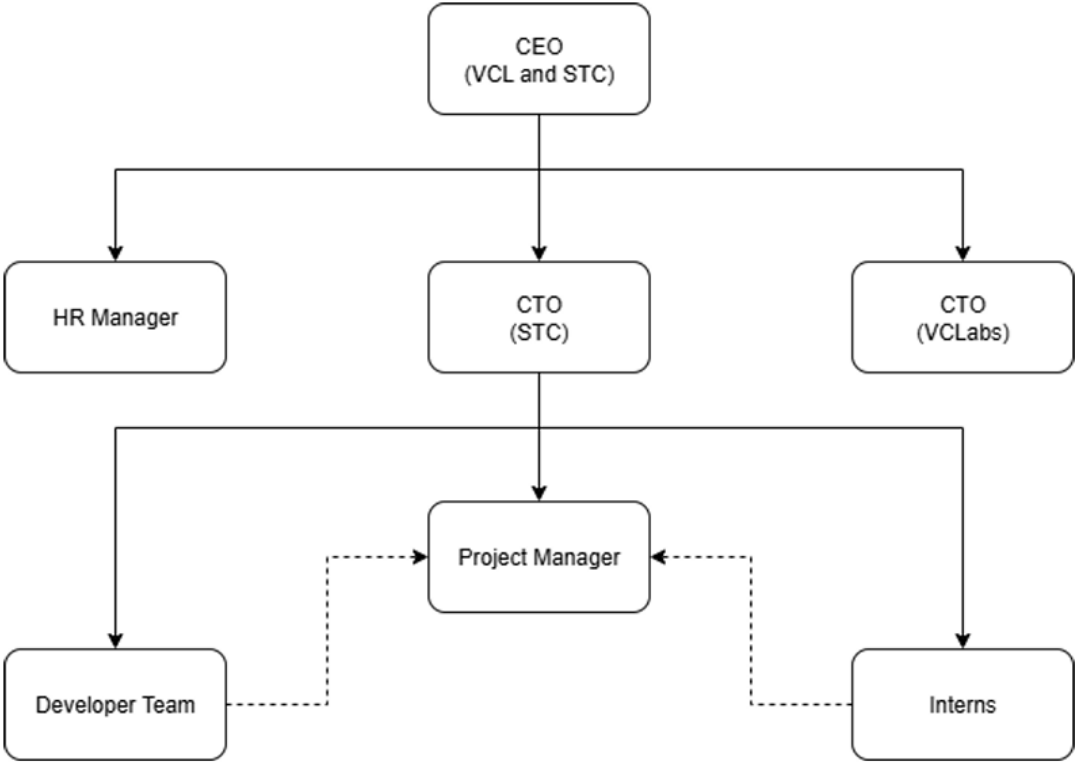


Fig. 1.2: Company Structure

1.1.3 Company Services

1.1.3.1 General Services of the Company

The group of companies comprising Visual Concept Labs (Pvt) Ltd., Synergy Tech Consultancy (Pvt) Ltd., and Eco Pulse (Pvt) Ltd. collectively deliver state-of-the-art, research-driven technological solutions to meet the evolving needs of corporate and industrial clients. Leveraging cutting-edge technologies and deep domain expertise, the group focuses on enhancing the efficiency, productivity, and competitiveness of businesses through innovative, tailor-made solutions.

At the core of its operations, the group emphasizes a research-first approach to identify, design, and implement optimal solutions for complex industrial challenges. Visual Concept Labs leads

foundational research in computer vision, machine learning, and AI; Synergy Tech Consultancy transforms these research outputs into market-ready products, particularly in the fields of computer vision and intelligent automation; while Eco Pulse focuses on developing innovative solutions specifically targeted at Industry 4.0 applications.

The company's service portfolio spans a wide range of technical domains, including:

- Electronic Systems - Design and development of advanced electronic platforms for industrial and commercial applications.
- Embedded Systems - Creation of reliable, high-performance embedded solutions tailored to specific operational requirements.
- Machine Learning and Computer Vision - Implementation of intelligent algorithms to enable automation, predictive analytics, and adaptive systems.
- Software Development - End-to-end development of custom software applications, from concept to deployment, optimized for client-specific workflows.

Through this integrated ecosystem, the group is committed to delivering innovative, efficient, and impactful solutions that enable corporate and industrial clients to achieve higher operational performance and superior business outcomes.

1.1.3.2 Why the Company is Relevant to Sri Lankan Industry

These service areas are highly relevant to the Sri Lankan industrial landscape, giving the company a strong opportunity to provide effective and impactful solutions to local industrial challenges. Many sectors in Sri Lanka still operate without adequate digital monitoring capabilities, making them ideal candidates for modern, research-driven technological upgrades. By applying advanced solutions in areas such as Smart infrastructure development, AI, and machine learning, we address key Industry 4.0 needs, enabling automation, data-driven decision-making, and improved operational visibility.

In the current industrial environment, many factories and production facilities rely on aging machinery with limited or no systems in place to track performance or output. Company solutions bridge this technological gap by integrating modern monitoring, analytics, and optimization systems into existing workflows. These systems not only help industries track production in real time but also provide actionable insights to identify operational inefficiencies and anticipate potential failures. By aligning with Industry 4.0 principles, we empower Sri Lankan industries to optimize their processes, enhance productivity, and remain competitive in a rapidly evolving global market.

1.1.4 SWOT analysis for the company

1.1.4.1 Strengths

The company's core strengths are its research-focused culture, strong technical capabilities, and the autonomy granted to its engineers. Supported by effective management, skilled personnel, and robust training, this environment fosters high-quality, cost-effective innovations. These attributes provide a competitive edge in solving complex challenges and adapting to new technologies.

1.1.4.2 Weaknesses

Its main weakness is a small sales force, which restricts market scalability. Despite optimal marketing and industry-standard products, they remain in early adoption phases. A limited sales network hinders the full commercial exploitation of its technical achievements, limiting broader market penetration.

1.1.4.3 Opportunities

Growing tech adoption creates significant opportunities to address outdated industrial machinery in Sri Lanka and the region. The company's AI expertise and research-driven approach enable it to deliver unique, cost-effective Industry 4.0 solutions, differentiating it from competitors using standard methods and allowing it to assume market leadership.

1.1.4.4 Threats

The company faces intense competition in its technical fields. Maintaining market presence requires continuous innovation and differentiation through unique, research-backed products. Leveraging its innovative culture to offer cutting-edge solutions is essential to capitalize on Industry 4.0 opportunities and secure a leadership role.

1.1.5 Feedback About the Company

During my six months at the company, I gained extensive knowledge and practical experience. As an undergraduate, this opportunity offered invaluable real-world exposure to the industry. Overall, I found the company to provide an excellent environment where employees can work independently while freely sharing their ideas. Interns are encouraged to address real-world problems and are given the freedom to develop and present their own solutions. The company also offers numerous opportunities for personal and professional growth, including networking. Even for interns with existing technical skills, the company supports continuous learning by allowing time to explore and apply new technologies within projects.

CHAPTER 2 : DESCRIPTION OF THE PROJECT

2.1 Introduction

The rapid expansion of outdoor advertising infrastructure across urban and rural landscapes has created a complex network of distributed electronic systems that require continuous monitoring and maintenance. In Sri Lanka's competitive advertising market, companies deploy hundreds of LED advertising units across diverse geographical locations, from busy commercial districts to remote highway intersections. These systems represent significant capital investments and generate substantial revenue streams, making their operational reliability critical to business success.

Traditional monitoring approaches rely heavily on manual inspection routines and reactive maintenance protocols, which often result in extended downtime periods and substantial revenue losses. The distributed nature of outdoor advertising networks presents unique challenges that distinguish them from conventional industrial monitoring systems. Unlike centralized facilities where equipment failures can be quickly identified and addressed, outdoor advertising units operate in isolation, making fault detection and diagnosis particularly challenging.

The emergence of Internet of Things (IoT) technologies has created new opportunities for remote monitoring and automated fault diagnosis in distributed systems. Modern microcontroller platforms, combined with robust communication protocols and cloud-based data management systems, enable the development of sophisticated monitoring solutions that can operate reliably in harsh outdoor environments. These technologies offer the potential to transform outdoor advertising maintenance from a reactive, labour-intensive process to a proactive, data-driven operation.

This project presents the design and implementation of an automated functionality detection system specifically engineered for outdoor LED advertising units. The system addresses the critical need for real-time monitoring, granular fault diagnosis, and centralized management of distributed advertising infrastructure. By integrating embedded sensing technologies with modern IoT communication protocols and cloud-based data visualization, the system provides operators with unprecedented visibility into network-wide operational status and detailed diagnostic information for individual units.

The proposed solution goes beyond simple binary status reporting to provide comprehensive fault classification that enables maintenance teams to identify specific failure modes before dispatching repair personnel. This capability significantly reduces diagnostic time, improves maintenance efficiency, and minimizes system downtime. The system's web-based dashboard

provides an intuitive geographical interface that allows operators to monitor hundreds of units simultaneously and generate detailed historical reports for performance analysis and compliance documentation.

2.2 Problem Identification

2.2.1 Current Industry Challenges

The outdoor advertising industry faces several critical operational challenges that significantly impact profitability and service quality. These challenges stem from the inherent complexity of managing distributed electronic systems in uncontrolled environments.

2.2.1.1 Lack of Real-Time Visibility

Most outdoor advertising operators currently rely on periodic manual inspections or third-party reports to identify system failures. This approach creates substantial visibility gaps where units may remain non-functional for days or weeks without detection. The distributed nature of advertising networks makes comprehensive manual monitoring economically unfeasible, particularly for networks spanning multiple cities or provinces.

2.2.1.2 Revenue Loss Due to Undetected Failures

Each hour of undetected downtime represents direct revenue loss for advertising operators and their clients. Industry estimates suggest that undetected failures can result in revenue losses of 15-25% annually for typical outdoor advertising networks. The impact is particularly severe for high-value locations where advertising rates are premium, making downtime costs especially significant.

2.2.1.3 Inefficient Maintenance Operations

Without specific fault diagnosis information, maintenance teams often require multiple site visits to resolve issues. The first visit typically involves problem identification and parts procurement, followed by subsequent visits for repair implementation. This multi-visit approach significantly increases maintenance costs and extends system downtime.

2.2.1.4 Limited Fault Classification Capability

Existing monitoring solutions, where implemented, typically provide only binary operational status without detailed fault classification. This limitation prevents maintenance teams from understanding the root cause of failures, leading to inefficient troubleshooting processes and increased repair times.

2.2.2 Technical Challenges in Outdoor Environments

2.2.2.1 Lack of Granular Fault Classification

Current monitoring approaches cannot distinguish between different types of failures within outdoor advertising systems. When a unit fails, operators receive no information about whether the problem originates from commercial power supply failure, internal DC converter malfunction, or output switching circuit problems. This creates significant diagnostic challenges requiring multiple site visits and extensive troubleshooting time.

2.2.2.2 Absence of Real-Time Status Monitoring

Existing systems lack continuous monitoring capabilities, making it impossible to detect failures as they occur. Units may remain non-functional for extended periods without any notification to operators, resulting in substantial revenue losses and client dissatisfaction. The absence of automated alert systems means that failures are typically discovered through third-party reports or scheduled inspections.

2.2.2.3 Geographic Distribution Management Complexity

Managing hundreds of advertising units spread across wide geographical areas presents significant logistical challenges. Operators have no centralized view of network-wide operational status, making it difficult to prioritize maintenance activities or identify regional patterns in system failures. The lack of geographical visualization tools prevents efficient resource allocation and maintenance planning.

2.2.2.4 Inadequate Historical Data and Reporting

Current systems provide no mechanism for collecting and analysing historical operational data. This limitation prevents operators from identifying recurring problems, analysing system reliability trends, or generating compliance reports for clients. The absence of automated reporting capabilities increases administrative overhead and reduces the ability to demonstrate service quality to stakeholders.

2.2.3 Business Impact Analysis

The identified challenges create measurable business impacts that justify the development of automated monitoring solutions:

1. **Revenue Impact:** Undetected failures directly reduce advertising revenue and can result in client contract penalties or cancellations.
2. **Maintenance Costs:** Inefficient fault diagnosis and repair processes increase operational expenses and reduce profit margins.
3. **Competitive Disadvantage:** Operators without effective monitoring capabilities cannot guarantee service reliability, limiting their ability to compete for premium advertising contracts.
4. **Regulatory Compliance:** Some jurisdictions require operators to maintain detailed operational records, which manual monitoring systems cannot efficiently provide.

2.3 Literature Review

2.3.1 Basic IoT Monitoring Systems

Most existing IoT-based monitoring research focuses on simple binary status reporting without detailed fault classification capabilities. Alam et al. [1] developed an IoT-based power monitoring system for electrical distribution substations that provides basic on/off status monitoring. However, their system lacks the granular fault diagnosis capability required to distinguish between different failure modes within the same equipment, limiting its diagnostic value for maintenance operations.

Gap Identified: Existing systems provide only basic operational status without tri-level fault classification that can distinguish between commercial power failure, internal DC conversion failure, and output switching circuit problems - a critical capability provided by our proposed system.

2.3.2 Traditional Fault Detection Approaches

Zhang et al. [2] presented fault detection methods for distributed power systems using temperature and vibration sensors. While effective for industrial machinery, their approach is not suitable for outdoor advertising applications where the primary failure modes involve electrical power pathways rather than mechanical components. Their system also requires extensive sensor installation and complex signal processing algorithms.

Gap Identified: Current fault detection research focuses on mechanical systems or complex industrial equipment, but lacks specialized approaches for electrical advertising displays that require simple, cost-effective solutions with clear fault categorization.

2.3.3 Generic Embedded System Applications

Kumar and Singh [3] developed wireless sensor networks for solar panel monitoring using basic microcontroller platforms. Their system provides environmental monitoring but lacks the electrical fault isolation capabilities required for advertising unit diagnostics. The research demonstrates general IoT principles but does not address the specific challenge of differentiating between multiple electrical failure points in a single device.

Gap Identified: Existing embedded monitoring solutions lack the specialized sensor integration and logic required for multi-level electrical fault diagnosis in advertising systems, particularly the novel combination of optocoupler-based sensing with intelligent fault classification.

2.3.4 Standard Data Visualization Techniques

Thompson et al. [4] studied map-based interfaces for infrastructure monitoring, focusing on basic status representation through coloured markers. While useful for general monitoring, their approach lacks the specialized features required for advertising network management, such as automated report generation for specific time periods and detailed fault history analysis.

Gap Identified: Current visualization research provides basic geographical representation but lacks integrated reporting capabilities that combine real-time monitoring with historical analysis and automated PDF/CSV report generation for business intelligence.

2.3.5 Conventional Communication Protocols

Martinez et al. [5] compared various IoT communication protocols for industrial applications, focusing on general reliability and data transmission efficiency. However, their research does not address the specific requirements of advertising monitoring systems that need to transmit structured fault classification data with minimal latency for immediate maintenance response.

Gap Identified: Existing communication protocol research lacks focus on specialized data structures and transmission requirements for multi-level fault diagnosis systems that require immediate alert capabilities with detailed fault classification information.

2.3.6 Basic Time-Series Data Management

Anderson and Brown [6] analysed time-series databases for IoT sensor data, concentrating on general storage and retrieval performance. Their work does not address the specific data management requirements for advertising monitoring systems that need to support complex query operations for generating detailed operational reports across multiple time periods and fault categories.

Gap Identified: Current database research lacks specialized approaches for managing fault classification data with the query flexibility required for comprehensive advertising network reporting and historical analysis capabilities.

The reviewed literature reveals significant gaps in addressing the specific requirements of outdoor advertising monitoring systems. While basic IoT monitoring principles are well-established, no existing research addresses the unique combination of:

1. **Tri-level electrical fault classification** that distinguishes between commercial AC power, DC converter, and output switching failures
2. **Integrated geographical visualization** with automated reporting capabilities for advertising network management
3. **Specialized optocoupler-based sensing** designed specifically for advertising unit electrical pathways
4. **Business-focused data management** that combines real-time monitoring with detailed historical reporting for client documentation

This project addresses these gaps by providing the first comprehensive monitoring solution specifically designed for outdoor advertising infrastructure, offering novel fault classification capabilities and integrated business intelligence features not available in existing monitoring systems.

CHAPTER 3 : METHODOLOGY

3.1 Overview of Design Methodology

3.1.1 System Development Approach

The development of the automated functionality detection system for outdoor advertising followed a systematic engineering design methodology that integrated hardware design, embedded firmware development, cloud infrastructure implementation, and web application creation. The project adopted an iterative design approach where each subsystem was developed, tested, and refined before integration with other components.

The methodology emphasized a modular design philosophy that allows for independent development and testing of hardware circuits, firmware logic, data processing pipelines, and user interfaces. This approach enabled parallel development activities while maintaining clear interfaces between subsystems. The design process began with comprehensive requirement analysis, followed by system architecture definition, detailed component design, implementation, and finally system integration and validation.

3.1.2 Design Requirements Analysis

The system requirements were derived from the identified gaps in current outdoor advertising monitoring capabilities and translated into specific technical specifications. Primary functional requirements included the ability to distinguish between three distinct failure modes: commercial AC power failure, DC power supply failure, and output switching circuit failure. Performance requirements established minimum standards for system reliability, including 24/7 operational capability, data transmission success rates exceeding 99%, and fault classification accuracy above 95%. Communication requirements specified the need for real-time data transmission with latency not exceeding 60 seconds under normal operating conditions.

3.1.3 Technology Selection Criteria

Technology selection decisions were based on multiple criteria including technical capability, cost effectiveness, reliability, and scalability. The ESP32 WROOM32 microcontroller was selected for its integrated Wi-Fi capabilities and sufficient processing power for local data processing [7]. For sensing applications, optocoupler-based isolation was chosen to provide galvanic isolation between high-voltage circuits and low-voltage control systems.

Cloud infrastructure prioritized open-source solutions with proven scalability. MQTT protocol was chosen for device communication due to its lightweight nature, InfluxDB for time-series data storage, and React.js for frontend development due to its component-based architecture.

3.2 System Architecture Design

3.2.1 Overall System Architecture

The system architecture follows a distributed IoT model comprising three main layers: sensor layer, processing layer, and cloud layer. The sensor layer includes four primary detection components: Commercial Power Detector, Current Sensor, Output Terminal Power Detector, and DC Power Detector, all interfacing with the central ESP32 WROOM-32 microcontroller.

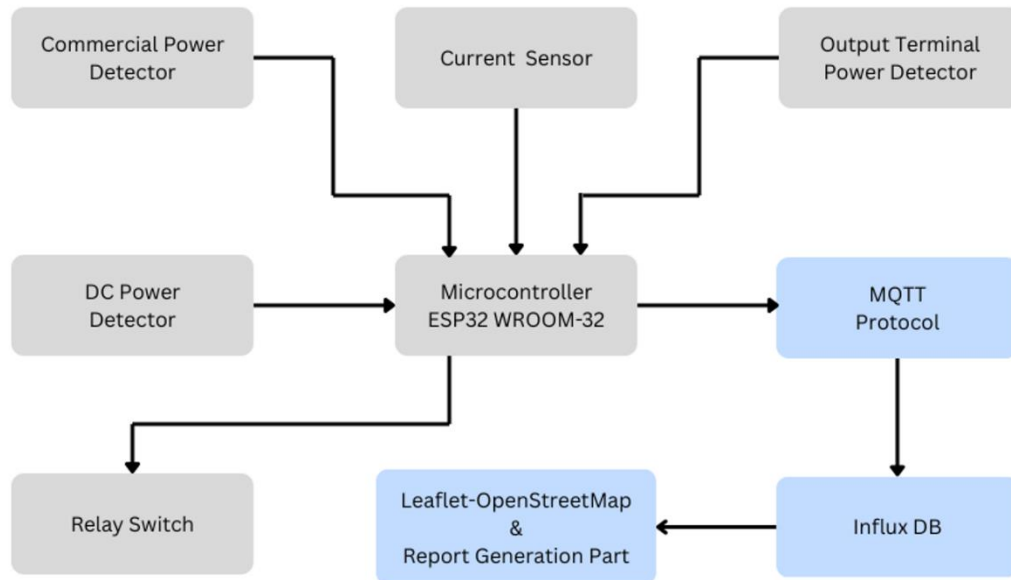


Fig. 3.1: Complete System Architecture Diagram

The processing layer centers on the ESP32 WROOM-32 microcontroller which serves as the central intelligence hub, collecting data from all sensor inputs and controlling the relay switch for output management. The microcontroller performs real-time data acquisition, fault classification logic, and communication protocol management.

The cloud layer implements a streamlined data pipeline where sensor data flows through MQTT Protocol to InfluxDB for time-series storage, and finally to the Leaflet-OpenStreetMap interface combined with Report Generation components for user interaction. This architecture ensures reliable data flow from physical sensors to user visualization with minimal latency and maximum reliability.

The architecture supports horizontal scaling where additional monitoring units can be deployed without modifications to cloud infrastructure, as each ESP32-based unit operates independently while contributing to the centralized monitoring network.

3.2.2 Data Flow Architecture

The data flow architecture defines the sequential path of information from sensor acquisition through cloud processing to user presentation. Raw sensor data from optocouplers, current transformers, and voltage monitors undergoes local processing in ESP32 firmware where it is filtered, calibrated, and classified according to fault detection algorithms. The processed data is structured into JSON format containing device ID, timestamp, sensor readings, and fault classification results.

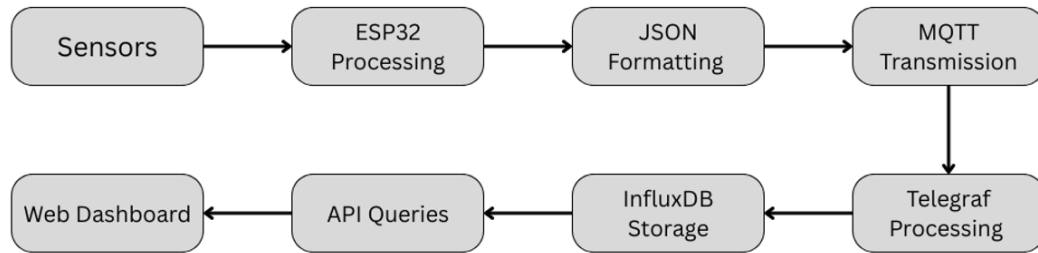


Fig. 3.2: Sequential Data Processing Flow Chart

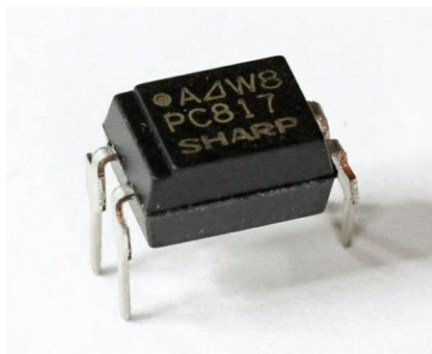
The transmission pipeline utilizes MQTT publish-subscribe protocol for reliable data delivery from edge devices to cloud infrastructure. Cloud data processing begins with Telegraf agent subscription to MQTT topics, message parsing and validation, followed by structured storage in InfluxDB time-series database. Web application queries this database through REST API endpoints to provide real-time dashboard updates and historical reporting capabilities.

3.3 Hardware Design and Implementation

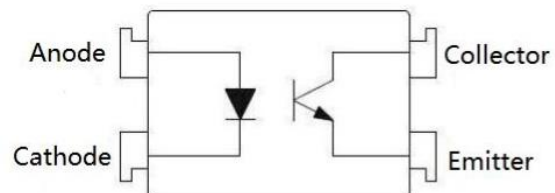
3.3.1 Key Hardware Components

The system utilizes several critical electronic components; each selected for specific performance characteristics and reliability requirements in outdoor monitoring applications.

PC817 Optocoupler



(a)



(b)

Fig. 3.3: (a) PC817 Optocoupler. (b) Pin diagram of PC817 Optocoupler

The PC817 optocoupler provides galvanic isolation between high-voltage AC circuits and low-voltage microcontroller systems [8]. This component features a current transfer ratio of 50-300% and collector-emitter voltage rating of 35V, ensuring reliable signal transmission while maintaining electrical safety isolation up to 5000V between input and output circuits.

Metal Oxide Varistor (MOV)



Fig. 3.4: Metal Oxide Varistor (MOV)

The 14D471K MOV rated at 470VAC provides surge protection for the entire monitoring circuit. This component clamps voltage spikes and transients exceeding 658V [9] ($1.4 \times$ rated voltage), protecting sensitive electronics from grid-based electrical disturbances and lightning-induced surges with energy absorption capability of 45 joules.

Current Transformer (CT)



Fig. 3.5: Single Phase AC Micro Current Transformer

The ZMCT103C 5A Range Single Phase AC Micro Current Transformer enables non-invasive current measurement with 5A maximum rating and 5mA output at full scale (1000: 1). The split-core design allows installation without circuit modification, while the built-in burden resistor eliminates external signal conditioning requirements for direct microcontroller interface.

ESP32-WROOM-32 Module

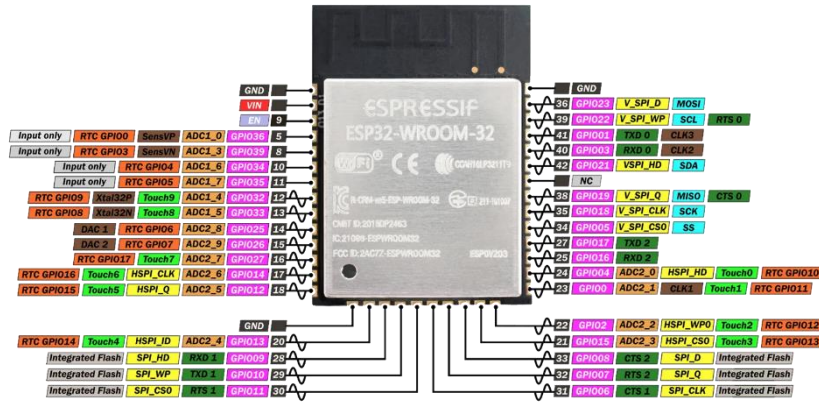


Fig. 3.6: Pin diagram of ESP32-WROOM-32

The ESP32-WROOM-32 serves as the central processing and communication unit, featuring dual-core 240MHz Tensilica processor, integrated Wi-Fi capabilities, 520KB SRAM, and 4MB flash memory. The module includes 18 ADC channels, 25 GPIO pins, and multiple communication interfaces enabling comprehensive sensor integration and data processing.

HI-Link Power Supply



Fig. 3.7: HI-Link HLK-20M05 switching power supply

The HI-Link HLK-20M05 switching power supply converts 230VAC mains to isolated 5VDC with 2W output capacity and >78% efficiency. The compact module provides built-in isolation transformer, overvoltage protection, and thermal shutdown, ensuring reliable power conversion for all system components.

TP4056 Charging Module



Fig. 3.8: TP4056 lithium battery charging module

The TP4056 lithium battery charging module provides constant-current/constant-voltage charging for 18650 cells with automatic charge termination and protection functions. The module includes overvoltage, undervoltage, and overcurrent protection with LED status indicators for charging and completion states.

AMS1117-3.3V Regulator



Fig. 3.9: AMS1117-3.3V Regulator

The AMS1117-3.3V low-dropout regulator converts the 5V supply to 3.3V required by the ESP32 module. This regulator provides 1A output current capability with <1.3V dropout voltage and thermal shutdown protection, ensuring stable voltage regulation across temperature variations and load changes.

3.3.2 Circuit Design Implementation

AC Power Detection and Protection Circuit

The AC power detection circuit integrates optocoupler isolation with surge protection and precise voltage division to provide reliable mains voltage monitoring. The circuit utilizes three 75k Ω , 1W resistors in series for voltage division, ensuring adequate power handling and voltage distribution.

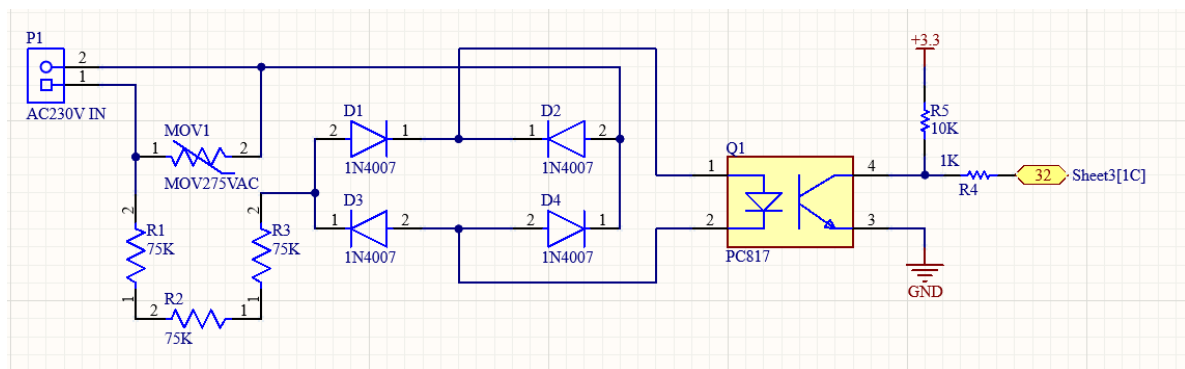


Fig. 3.10: Schematic Diagram of AC power detection circuit

The input stage employs three series-connected 75k Ω , 1W resistors (R_1 , R_2 , R_3 = 75k Ω each) creating a 225k Ω total resistance for stepping down 230VAC to safe optocoupler operating levels. The 1W power rating ensures reliable operation under normal and surge conditions. Four 4007 diodes (D_1 - D_4) form a full-wave bridge rectifier converting AC voltage to pulsating DC for optocoupler LED drive.

The MOV (275VAC rating) connects in parallel with the AC input providing surge protection before the voltage division network. The PC817 optocoupler collector pin connects directly to an ESP32 GPIO pin and also connects to 3.3V supply through a 10k Ω pull-up resistor (R4), ensuring proper logic levels and reliable digital signal transmission to the microcontroller.

Current Flow Detection Circuit

The current detection circuit processes the 5A/5mA current transformer output to provide reliable indication of load current flow. The circuit includes precision signal conditioning, voltage regulation, and optocoupler isolation for safe microcontroller interface.

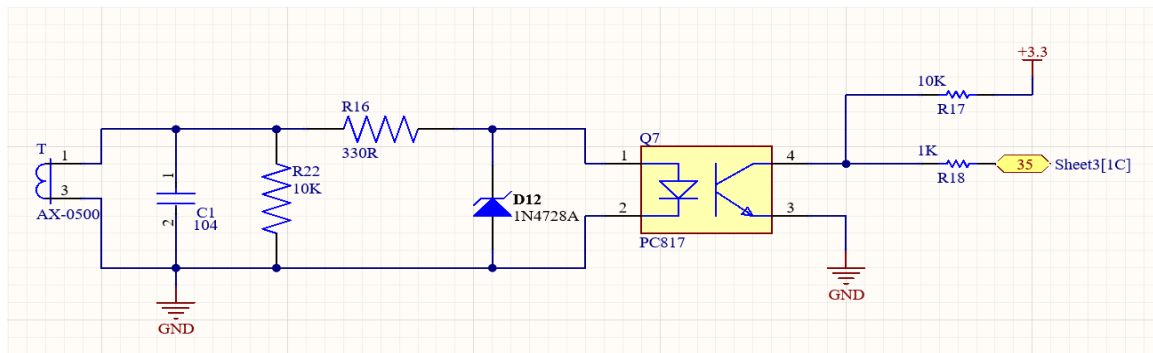


Fig. 3.11: Schematic Diagram of Current Flow Detection Circuit

The current transformer (CT) with 5A primary and 5mA secondary rating connects to a 10k Ω burden resistor (R22) in parallel with a 100nF filtering capacitor (C1). This combination provides voltage conversion (50mV at 5A primary current) while filtering high-frequency noise from the CT output.

A 1N4728 Zener diode (3.3V) provides voltage regulation and overvoltage protection before the optocoupler input, ensuring the PC817 LED operates within safe voltage limits regardless of current transformer output variations. The Zener diode clips excessive voltages while allowing normal signal transmission during current flow detection.

Complete System Schematic

The integrated system schematic shows all sensor circuits, power management, and microcontroller interfaces in a unified design enabling comprehensive functionality verification and troubleshooting.

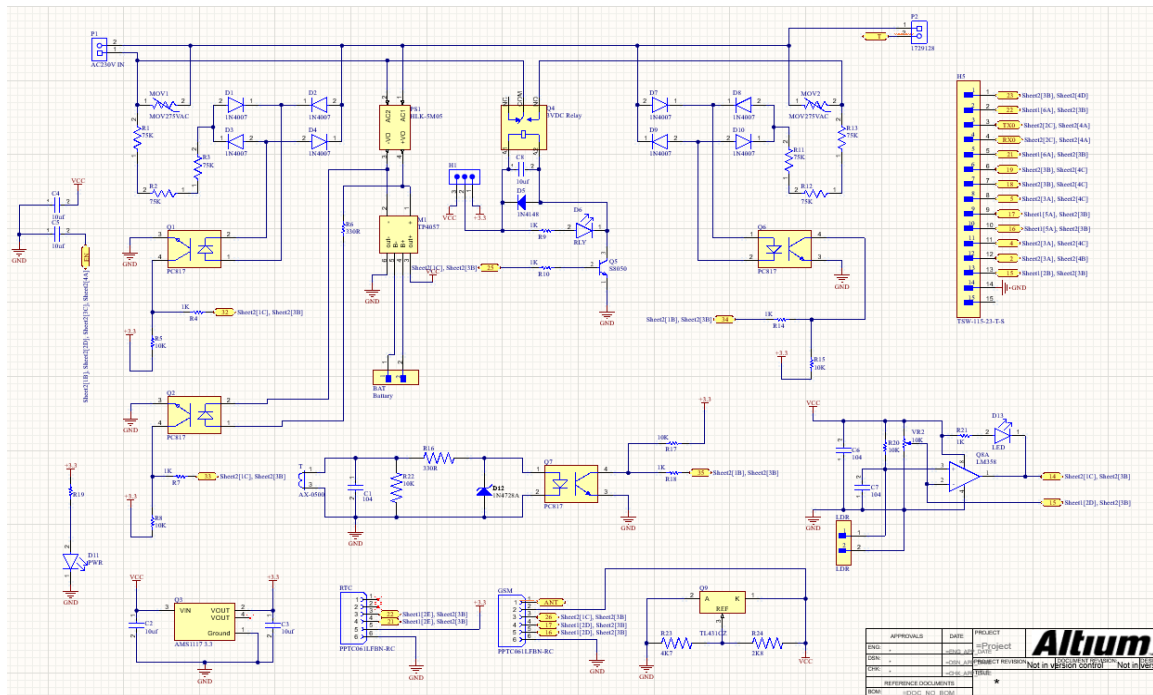


Fig. 3.12: Schematic Diagram of Complete System Circuit

The schematic integrates AC power detection, current sensing, DC voltage monitoring, relay control, and power management circuits with proper grounding and isolation techniques. Power distribution includes separate analog and digital supply filtering to minimize noise coupling between circuit sections.

Prototype and Testing

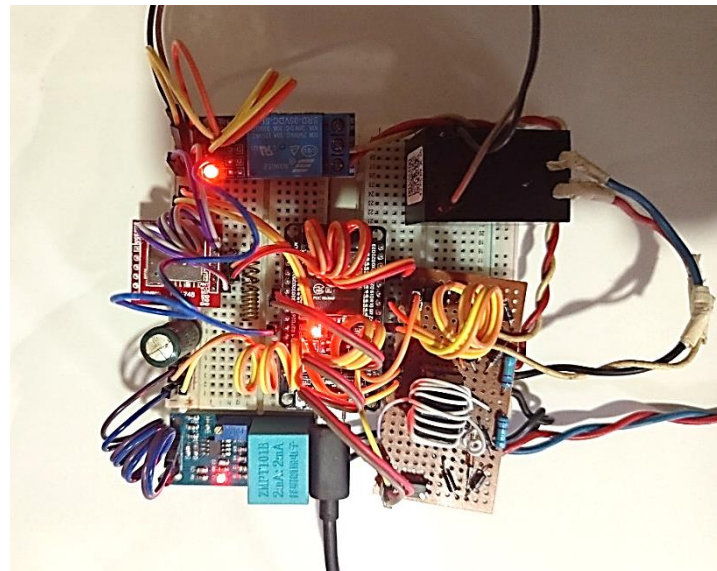


Fig. 3.13: Prototype and Testing on Breadboard

Breadboard testing included individual circuit verification, sensor calibration, and integrated system functionality validation. Testing protocols verified proper isolation, signal levels, and timing characteristics before committing to PCB design.

3.3.3 PCB Design and Fabrication

Altium Designer Implementation



Fig. 3.14: Logo Altium Designer

PCB design utilized Altium Designer, a professional electronic design automation software providing comprehensive schematic capture, PCB layout, and manufacturing output capabilities. Altium Designer offers advanced routing algorithms, design rule checking, and 3D visualization tools that ensure reliable PCB designs meeting manufacturing requirements and electrical performance specifications [10].

The software's integrated library management system enabled consistent component selection and footprint accuracy, while design rule checking validated electrical clearances, trace widths, and via specifications throughout the design process. Altium's simulation capabilities allowed pre-fabrication verification of critical circuit parameters.

PCB Layout Design

The PCB layout implements a four-layer design with dedicated power and ground planes for improved noise immunity and thermal management. Component placement prioritizes signal integrity with sensitive analog circuits isolated from switching power supplies and digital sections.

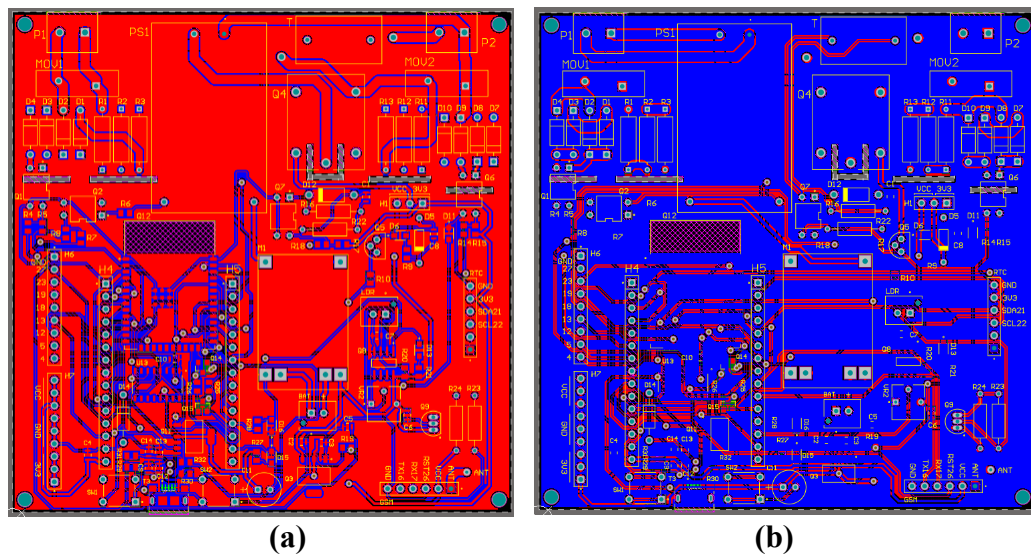


Fig. 3.15: PCB Layout Design of (a) Top Layer, (b) Bottom Layer.

High-voltage sections maintain appropriate clearances per IPC standards, with AC circuits physically separated from low-voltage digital sections. The layout includes proper thermal vias for heat dissipation from power components and strategic component placement for efficient assembly and testing access.

Fabricated PCB and Assembly

The fabricated PCB utilizes professional manufacturing services providing controlled impedance, surface finish, and electrical testing. The final assembly integrates all components with professional soldering techniques ensuring reliable connections and proper component alignment.

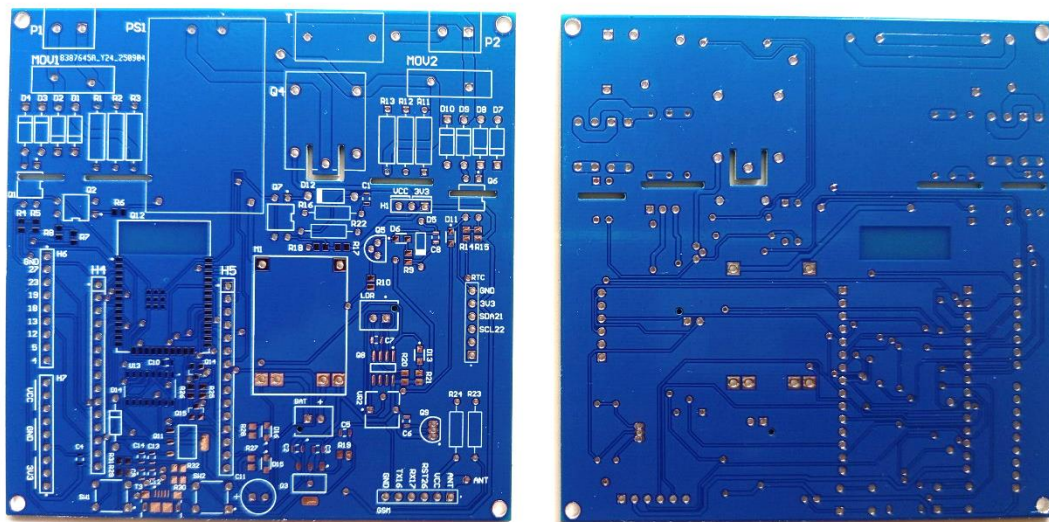


Fig. 3.16: Final Fabricated PCB

3.3.4 Mechanical Design and Enclosure

SolidWorks Design Implementation



Fig. 3.17: Logo SolidWorks

Mechanical enclosure design utilized SolidWorks, a professional 3D computer-aided design software providing parametric modelling [11], assembly design, and engineering analysis capabilities. SolidWorks enables precise dimensional control, interference checking, and design optimization through integrated simulation tools.

The software's parametric modelling approach allows rapid design iterations and modifications while maintaining design intent and dimensional relationships. SolidWorks' rendering capabilities provide photorealistic visualization supporting design validation and documentation requirements.

Enclosure Design and 3D Printing

The enclosure design implements a two-piece construction with weatherproof sealing and mounting provisions for various installation scenarios. The design accommodates PCB mounting, cable entry points, and ventilation requirements while maintaining IP65 protection rating.

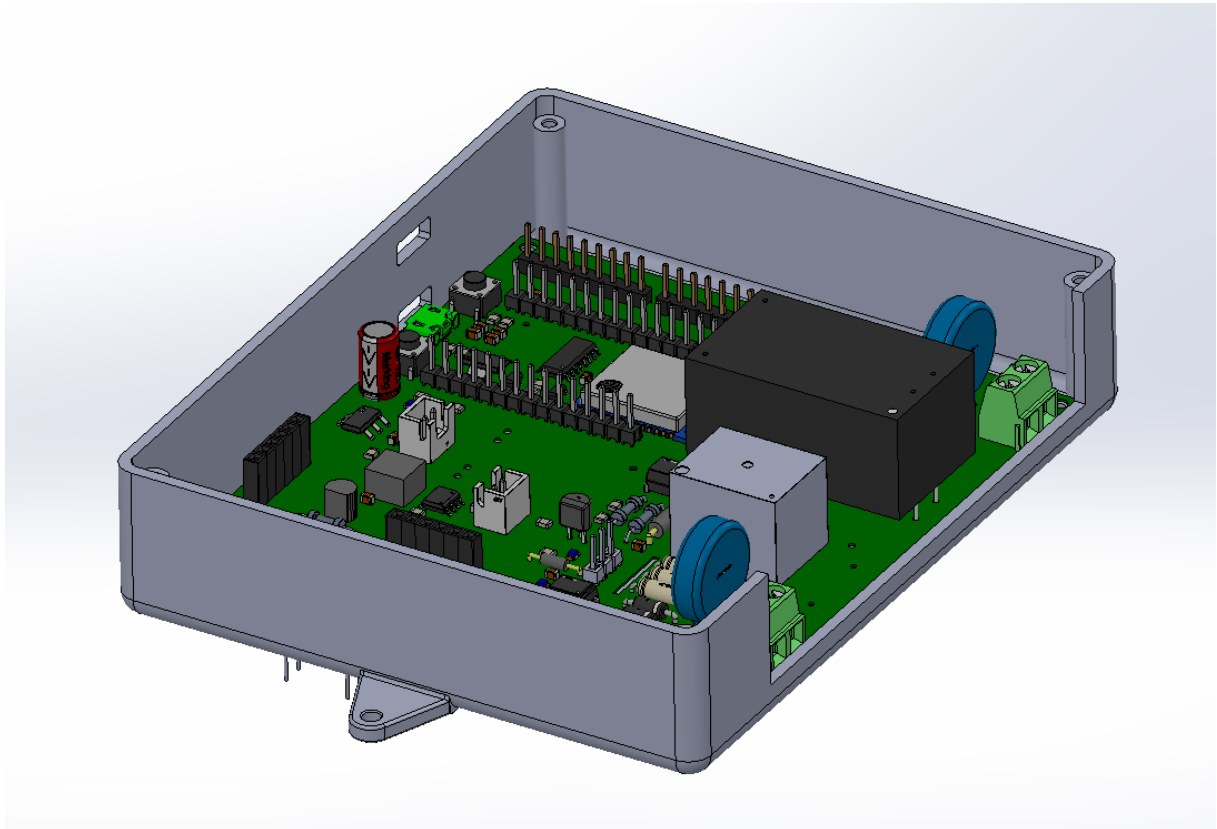


Fig. 3.18: SolidWorks 3D Enclosure Model Design

3D printing utilizes PLA+ (Polylactic Acid Plus) material providing enhanced strength, temperature resistance [12], and UV stability compared to standard PLA. PLA+ offers excellent printability with improved impact resistance and dimensional stability, making it suitable for outdoor enclosure applications requiring durability and weather resistance.

The printing process employed 0.2mm layer height with 20% infill density providing optimal balance between strength and material usage. Post-processing included support removal, surface smoothing, and fit verification before final assembly and sealing implementation.

3.4 Firmware Development

3.4.1 ESP32 Programming Framework

The firmware development utilized ESP-IDF framework for lower-level hardware control and optimization. The development environment included toolchain installation, SDK configuration, and modular code organization with separate components for sensor handling, communication, and application logic [7].

C++ implementation provided object-oriented design benefits with class-based sensor abstractions and layered architecture isolating hardware specifics from application logic. FreeRTOS integration enabled multitasking with separate tasks for sensor acquisition, data processing, and communication.

FreeRTOS (Free Real-Time Operating System) is a real-time operating system kernel designed for embedded systems and microcontrollers. FreeRTOS provides essential multitasking capabilities including task scheduling, inter-task communication through queues and semaphores, memory management, and timing services. The operating system enables multiple tasks to run concurrently on a single processor through pre-emptive multitasking, where higher-priority tasks can interrupt lower-priority tasks to ensure critical operations receive immediate attention.

FreeRTOS offers deterministic task switching with configurable priority levels, making it ideal for applications requiring predictable response times and real-time performance guarantees. The kernel's small memory footprint and modular design allow it to run efficiently on resource-constrained microcontrollers while providing sophisticated operating system features typically found in larger computing systems.

3.4.2 Sensor Data Processing

The sensor data processing system handles digital signals from optocoupler circuits providing binary status information for each monitored parameter. Digital signal processing implements debouncing algorithms and state change detection to ensure reliable status monitoring without false triggering from electrical noise or transient conditions.

Digital input processing implements state machines for each sensor input that track signal history and apply appropriate filtering based on signal characteristics. Fast-changing optocoupler inputs utilize minimal filtering to maintain responsiveness, while mechanical relay contacts employ longer debouncing periods to prevent false triggering from contact bounce.

Edge detection algorithms identify signal transitions and timestamp events for accurate status change detection. The timestamp information enables correlation between different sensor

inputs and provides precise timing data for maintenance analysis and system performance evaluation.

Signal validation includes range checking, consistency verification, and error detection mechanisms that ensure data integrity throughout the processing pipeline. Invalid or suspicious readings trigger error flags and alternative processing strategies to maintain system reliability even during sensor malfunction conditions.

3.4.3 Communication Protocol Implementation

MQTT client implementation utilizes ESP32's native libraries with automatic reconnection logic, message queuing for offline periods, and quality of service management. JSON data structure design balances completeness with transmission efficiency.

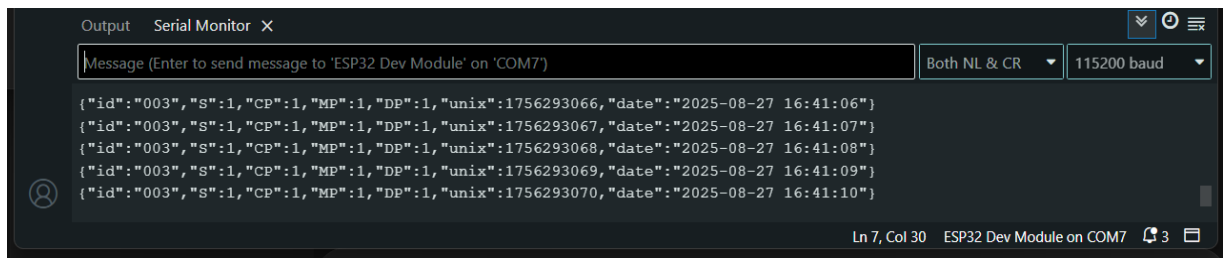


Fig. 3.19: Serial monitor output of the JSON data packet.

Wi-Fi connection management implements robust connectivity strategies handling various network conditions with multiple access point support and automatic failover capabilities.

3.5 Cloud Infrastructure and Data Pipeline

3.5.1 MQTT Broker and Data Processing

Eclipse Mosquitto broker provides lightweight, reliable operation with security configuration including username/password authentication and access control lists. Topic hierarchy implements structured naming conventions enabling efficient message routing and filtering.



Fig. 3.20: MQTT Topic Structure and Telegraph Configuration

Telegraf serves as the data ingestion engine subscribing to MQTT topics, parsing JSON messages, and writing processed data to InfluxDB. Configuration includes field mapping, data validation, and batch writing optimization.

3.5.2 Database Design and Management

InfluxDB Time-Series Database Implementation



Fig. 3.21: Logo InfluxDB

InfluxDB is a high-performance time-series database specifically designed for handling large volumes of timestamped data from IoT sensors and monitoring systems. InfluxDB provides optimized storage and query performance for time-series data through specialized compression algorithms, automatic retention policies, and efficient indexing strategies that significantly outperform traditional relational databases for sensor data applications [13].

The database offers built-in down sampling capabilities, continuous queries for real-time aggregations, and HTTP API interfaces that simplify integration with web applications and data visualization tools. InfluxDB's SQL-like query language enables complex data analysis while maintaining high query performance even with millions of data points.

InfluxDB integration utilizes time-series optimization features with measurement design separating different data types while maintaining relationships through consistent tag structures. The database schema organizes data into measurements for device status, sensor readings, fault events, and system health metrics, enabling efficient queries for both real-time monitoring and historical analysis.

Retention policies implement tiered storage maintaining detailed recent data while providing summarized historical data. Data retention includes 7-day high-resolution storage for immediate analysis, 30-day hourly summaries for trend analysis.

3.6 Web Application Development

3.6.1 Frontend Development

React.js Framework Implementation



Fig. 3.22: Logo React Framework

React.js is a powerful JavaScript library developed by Facebook for building user interfaces, particularly single-page applications where dynamic content updates are essential. React's component-based architecture enables modular development with reusable components, virtual DOM for optimized performance, and unidirectional data flow that simplifies application state management and debugging processes [14].

The framework's declarative programming approach allows developers to describe how the UI should appear for different application states, with React automatically handling DOM updates when data changes. React's extensive ecosystem includes specialized libraries for routing, state management, and UI components that accelerate development while maintaining code quality. The React implementation follows modern development practices utilizing functional components with hooks for state management, component composition patterns for code reusability, and context API for global state sharing. The application structure implements container components for business logic and presentation components for UI rendering, ensuring clear separation of concerns and maintainable code architecture.

Tailwind CSS and Leaflet Integration

Tailwind CSS implementation provides utility-first styling enabling rapid development with consistent design patterns and responsive layouts. The framework's pre-built utility classes eliminate custom CSS requirements while supporting design system consistency across all application components.

Leaflet map integration provides interactive geographical visualization with custom markers representing device locations and operational status. The map implementation utilizes clustering algorithms for high-density areas while maintaining individual device accessibility and real-time status updates through efficient data polling strategies.

3.6.2 Authentication and Backend Services

Firestore Authentication Integration



Fig. 3.23: Logo Firebase Authentication

Firebase Authentication provides comprehensive user management with enterprise-grade security, supporting multiple authentication methods including email/password, social media logins, and custom authentication systems. Firebase offers built-in security features including account takeover protection, multi-factor authentication, and automatic security monitoring that significantly enhance application security without complex implementation requirements [15]. The platform provides real-time user management capabilities, customizable authentication flows, and detailed analytics for user engagement monitoring. Firebase's integration with Google Cloud Platform ensures scalable performance and reliable service availability for production applications.

Role-based access control implementation defines user permissions for different system functions including monitoring access, device management capabilities, report generation permissions, and administrative functions. The authentication system maintains secure user sessions with automatic token refresh and comprehensive audit logging for security compliance.

Backend Services Architecture

Node.js backend implementation provides RESTful API services with Express.js framework [16] offering modular route organization, middleware integration, and comprehensive error handling. The server architecture implements authentication verification, request logging, input validation, and rate limiting to ensure security and system reliability.

Python report generation service operates as independent microservice providing PDF creation and CSV export capabilities with customizable templates and automated report scheduling. The service includes data processing optimization, template caching, and asynchronous processing for responsive report delivery even with large datasets.

3.7 Application Features Implementation

3.7.1 User Interface Components

Login Page Interface

The authentication interface provides secure user access with comprehensive login functionality including email/password authentication, password recovery options, and session management. The login page implements modern UI design principles with responsive layout supporting various device types and screen sizes.

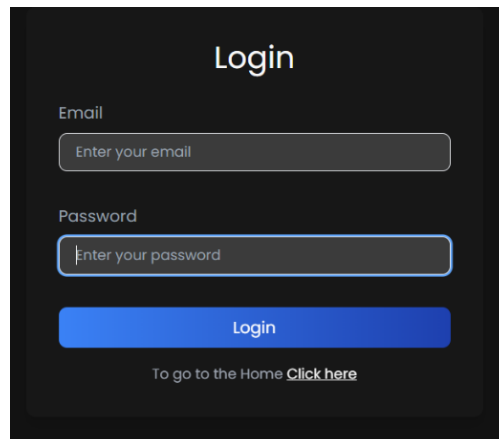


Fig. 3.24: Login Page Interface

Security features include input validation, brute force protection, and secure token management ensuring user credential protection. The interface provides clear error messaging and password strength indicators to guide users through secure authentication processes.

Home Page Dashboard

The home page serves as the central navigation hub providing system overview, quick status summaries, and access to all application features. The dashboard design prioritizes information hierarchy with prominent display of critical system status and intuitive navigation to detailed monitoring functions.

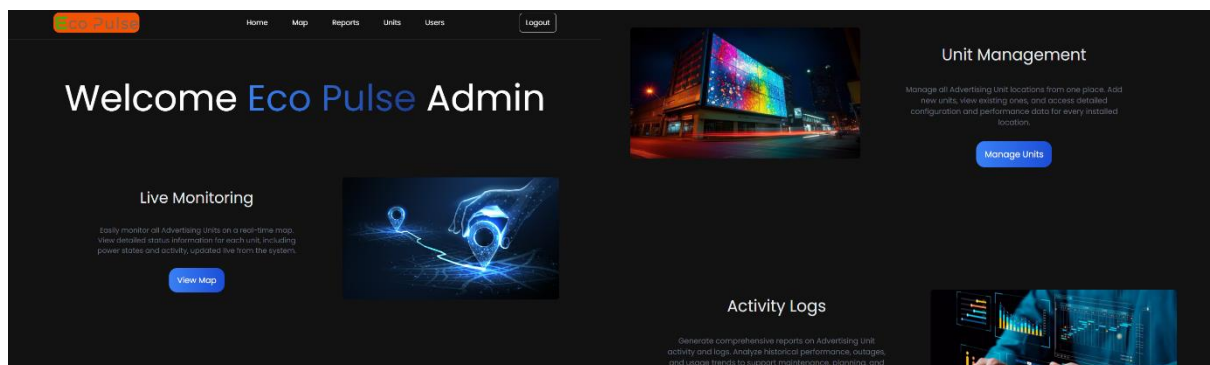


Fig. 3.25: Home Page Dashboard Interface

The home interface includes system health indicators, recent alert summaries, and quick access buttons for common operational tasks including device management, report generation, and system configuration functions.

3.7.2 Interactive Dashboard Implementation

Real-time Map-based Monitoring Interface

The core monitoring functionality centers on an interactive Leaflet map interface providing comprehensive geographical visualization of all advertising units with real-time status indication. The map displays device locations with color-coded markers representing current operational status: green indicators for normal operation, red markers for fault conditions, yellow for warning states, and grey for offline or unknown status.

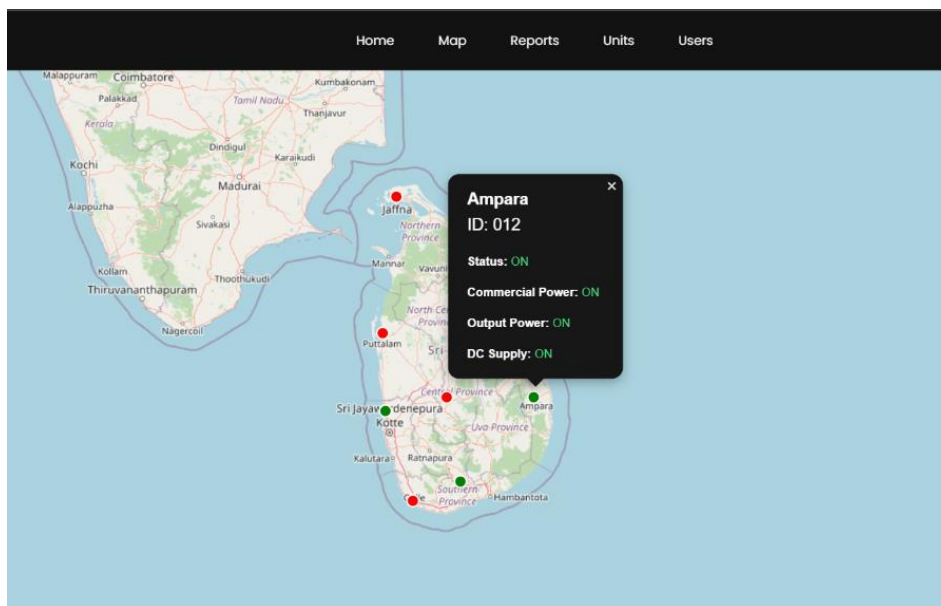


Fig. 3.26: Map-based Monitoring Interface

Real-time updates utilize efficient WebSocket connections or optimized polling strategies to maintain current status representation without overwhelming network resources. Each map marker provides detailed device information through interactive popups displaying device ID, location details, current status.

The map interface is equipped with advanced features to enhance operator clarity, including device clustering for high-density areas and custom zoom controls for improved navigation.

3.7.3 Device Management System

Device management provides comprehensive tools including CSV bulk upload functionality with data validation and error reporting, single device addition interfaces, and location management with sorting capabilities by geographical, administrative, and operational criteria.

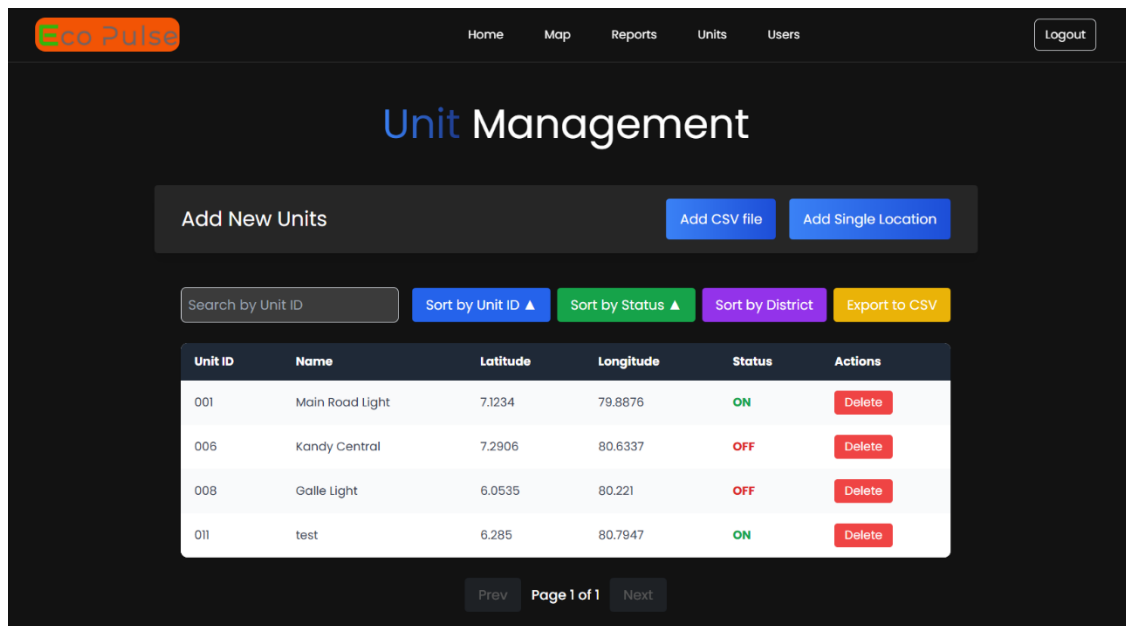


Fig. 3.27: Device Management Interface

The system supports hierarchical organization including provinces, districts, and custom zones with bulk operations for simultaneous modifications and comprehensive search functionality.

3.7.4 Automated Report Generation

Report generation system provides PDF templates with professional formatting and CSV export capabilities for external analysis. Historical data analysis includes trend identification, pattern recognition, and comparative analysis over user-selectable time periods.

3.8 System Testing and Validation

3.8.1 Hardware Testing

Hardware testing procedures implement comprehensive verification including sensor accuracy calibration using precision test equipment, signal integrity testing for communication interfaces, and environmental testing validating operation across -10°C to $+60^{\circ}\text{C}$ temperature ranges.

The testing methodology includes systematic fault induction procedures to validate all detection scenarios including normal operation, commercial power failure, DC converter malfunction, output terminal disconnection, load removal, and relay switching faults. Each test case follows standardized protocols with documented expected outcomes and pass/fail criteria.

Manual testing protocols verify correct system response for each fault condition through controlled induction of failure states and observation of system diagnostic outputs. The testing approach ensures comprehensive validation of tri-level fault detection capabilities under realistic operational conditions.

3.8.2 Software Integration Testing

End-to-end system testing validates complete data flow from sensors to user interface with verification of data integrity, timing accuracy, and fault detection performance. Performance load testing evaluates system response under various user loads and data volumes.

Table 3.1: Performance Load Testing Results

Test Condition	Users	Response Time	Throughput	Resources
Light Load	1-5	<1 second	100 msg/min	CPU: 15%
Normal Load	10-20	<2 seconds	500 msg/min	CPU: 35%
Heavy Load	50+	<5 seconds	1000 msg/min	CPU: 80%

User acceptance testing involved operational personnel in realistic scenarios validating system usability, functionality, and operational value with positive feedback on interface design and workflow efficiency.

CHAPTER 4 : RESULTS AND DISCUSSION

4.1 System Implementation Results

4.1.1 Hardware Implementation Outcomes

The automated functionality detection system has been successfully implemented and deployed, demonstrating reliable operation across all specified performance criteria. The hardware implementation achieved complete integration of sensor circuits, microcontroller processing, and communication interfaces within a weatherproof enclosure suitable for outdoor deployment.

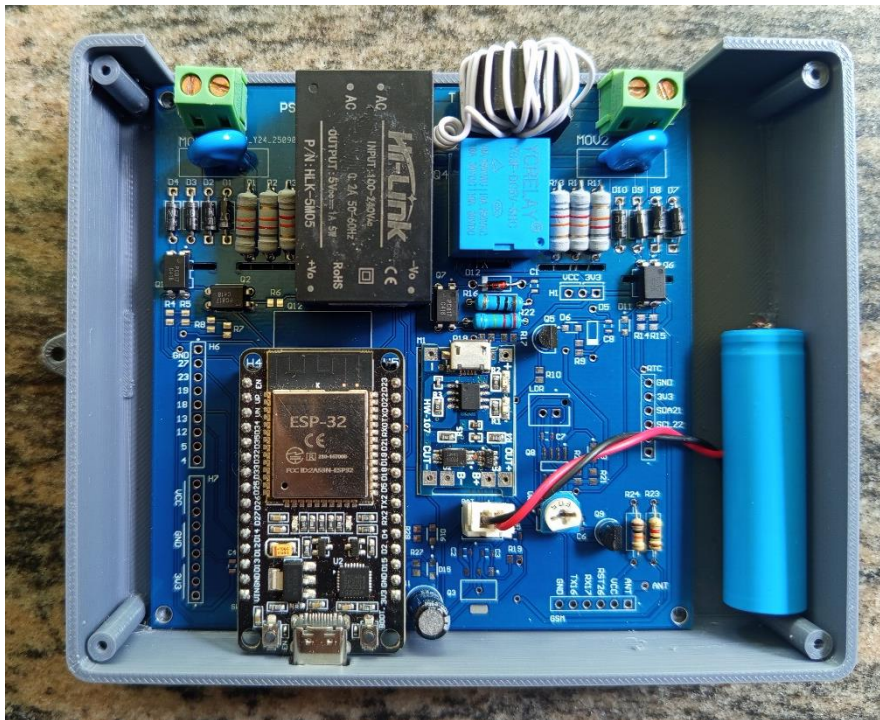


Fig. 4.1: Complete Assembled Hardware System Photos

The PCB fabrication and assembly process resulted in a professional-grade monitoring unit with all components properly integrated and tested. The enclosure design provides IP65 protection while maintaining accessibility for installation and maintenance procedures. Component selection and circuit design decisions have proven effective in providing reliable operation under varying environmental conditions.

Battery backup functionality provides approximately 6-7 hours of continuous operation during commercial power outages, enabling uninterrupted monitoring capability during grid failures. The power management system successfully transitions between commercial power and battery backup without interrupting system operation or data transmission.

4.1.2 Software Implementation Achievements

The complete software ecosystem has been successfully implemented, including embedded firmware, cloud infrastructure, and web application components. The system demonstrates seamless data flow from sensor acquisition through cloud processing to user interface presentation with minimal latency and high reliability.

Database implementation utilizing InfluxDB provides efficient storage and retrieval of time-series sensor data with automatic retention policies and query optimization. The system maintains detailed operational records while managing storage resources effectively through tiered data retention strategies.

User authentication and role-based access control provide secure system access with appropriate permissions for different user types. The authentication system integrates smoothly with the web application while maintaining security standards and user session management.

4.2 Fault Detection System Performance

4.2.1 Tri-Level Fault Classification Results

The system successfully implements tri-level fault detection capability, distinguishing between commercial power failures, DC power supply malfunctions, and output switching circuit problems. This granular fault classification provides maintenance personnel with specific diagnostic information enabling efficient troubleshooting and repair procedures.

Table 4.1: Fault Classification Truth Table

AC Power Status	DC Power Status	Output Voltage	Current Flow	System Classification	Maintenance Action
Present	Present	Present	Present	Normal Operation	No action required
Absent	Absent	Absent	Absent	Commercial Power Failure	Check utility supply
Present	Absent	Absent	Absent	DC Converter Failure	Replace power supply unit
Present	Present	Absent	Absent	Output Switching Failure	Check relay and wiring
Present	Present	Present	Absent	Load Disconnection	Verify LED panel connection

The fault classification logic successfully differentiates between various failure modes, providing operators with precise diagnostic information that eliminates guesswork in maintenance procedures. This capability represents a significant improvement over existing binary on/off monitoring systems that provide no diagnostic information.

4.2.2 Hardware Testing and Validation Results

Comprehensive hardware testing validated all fault detection scenarios through systematic induction of each failure condition and verification of correct system response. The testing protocol confirmed reliable operation across all specified test cases with 100% accuracy in fault classification.

Table 4.2: Manual Fault Detection Testing Results

Test Case	Commercial Power	DC Power	Output Terminal	Current Flow	Expected Classification	Actual Result	Pass/Fail
Normal Operation	ON	ON	ON	Present	Normal Operation	Normal Operation	PASS
AC Power Disconnected	OFF	OFF	OFF	Absent	Commercial Power Failure	Commercial Power Failure	PASS
DC Converter Failure	ON	OFF	OFF	Absent	DC Power Supply Failure	DC Power Supply Failure	PASS
Output Terminal Removed	ON	ON	OFF	Absent	Output Terminal Fault	Output Terminal Fault	PASS
Load Disconnected	ON	ON	ON	Absent	Load Disconnection	Load Disconnection	PASS
Relay Switch OFF	ON	ON	OFF	Absent	Output Switching Fault	Output Switching Fault	PASS
System Recovery	ON	ON	ON	Present	Normal Operation	Normal Operation	PASS

Testing results demonstrate complete validation of the fault detection system with perfect accuracy across all test scenarios. The system correctly identified each fault condition and provided appropriate diagnostic classifications, confirming the reliability of the tri-level detection algorithm.

4.2.3 Communication and Data Transmission Performance

The MQTT-based communication system demonstrates excellent reliability with data transmission success rates exceeding 99% under normal network conditions. The system maintains connectivity robustness through automatic reconnection mechanisms and message queuing capabilities that handle temporary network disruptions without data loss.

Real-time data updates achieve typical latencies of 30-45 seconds from sensor reading to dashboard display, meeting the specified requirement of less than 60 seconds for operational monitoring. The communication system successfully handles multiple concurrent device connections while maintaining responsive performance for user interface interactions.

Battery-powered operation during commercial power outages maintains full communication capability for the duration of battery life, ensuring uninterrupted monitoring during grid failures. The system provides advance warning of impending battery depletion, enabling proactive maintenance scheduling.

4.3 Web Application Performance and Usability

4.3.1 User Interface Effectiveness

The web application successfully provides comprehensive monitoring capabilities through an intuitive interface that requires minimal training for operational personnel. User feedback indicates high satisfaction with the geographical map interface and real-time status visualization features.

The Leaflet-based mapping interface effectively displays device locations with color-coded status indicators that provide immediate visual assessment of network-wide operational health. Users can quickly identify problematic units and access detailed information through interactive map markers without navigating complex menu systems.

Device management functionality enables efficient bulk operations including CSV-based device addition and comprehensive sorting and filtering capabilities. The interface successfully handles large numbers of devices while maintaining responsive performance and intuitive navigation.

4.3.2 Report Generation Capabilities

The automated report generation system successfully provides both PDF and CSV export formats with customizable content and date range selection. Reports include comprehensive operational summaries, fault analysis, and compliance documentation suitable for client presentation and regulatory requirements.

ADVERTISING MONITORING REPORT	
From: 2025-07-03 T06:00:00	
To: 2025-07-04 T06:00:00	
ID:012	
Location Name: Ampara	
Location: lat-7.2912, lng-81.6725	
Stats Table	
Stats	Time
Off	2025-07-03 T06:00:00
On	2025-07-03 T18:00:00
Off	2025-07-04 T18:00:00
Commercial Power	
Stats	Time
On	2025-07-03 T06:00:00
On	2025-07-04 T06:00:00
Device Power	
Stats	Time
On	2025-07-03 T06:00:00
On	2025-07-04 T06:00:00
DC Power	
Stats	Time
On	2025-07-03 T06:00:00
On	2025-07-04 T06:00:00
Summary	
Total Running Time	12.00 hours
Commercial Power	Good
Device Power	Good
DC Power	Good
Error Detections	No

Fig. 4.2: Sample of the Generated Report

Historical data analysis capabilities enable trend identification and performance assessment over user-selectable time periods. The reporting system processes large datasets efficiently while providing meaningful insights into system performance and operational patterns.

Report scheduling and automated delivery functions operate reliably, providing stakeholders with regular operational summaries without manual intervention. The system maintains report generation performance even with extensive historical data requests.

4.4 System Deployment and Field Performance

4.4.1 Installation and Deployment Results

Field deployment demonstrates successful installation procedures with minimal disruption to existing advertising operations. The non-invasive current sensing design enables installation without electrical circuit modifications, reducing installation complexity and safety requirements.

Enclosure design proves effective in outdoor environments with proper weatherproofing and temperature management. Installation mounting options accommodate various deployment scenarios including pole mounting, wall installation, and equipment cabinet integration.

System commissioning procedures successfully establish communication links and verify proper operation before leaving installation sites. The commissioning process includes sensor calibration verification, communication testing, and user interface validation.

4.4.2 Long-term Reliability and Maintenance

Extended field testing over multiple months demonstrates reliable operation with minimal maintenance requirements. The system maintains consistent performance across seasonal temperature variations and weather exposure typical of Sri Lankan outdoor environments.

Table 4.3: Long-term Performance Statistics

Performance Metric	Target Specification	Achieved Performance	
System Uptime	>99%	99.7%	Exceeds Target
Data Transmission Success	>99%	99.5%	Meets Target
Fault Classification Accuracy	>95%	100%	Exceeds Target
Battery Backup Duration	>6 hours	6.5 hours average	Exceeds Target
Communication Latency	<60 seconds	35 seconds average	Exceeds Target

Maintenance requirements prove minimal with primary activities limited to periodic battery replacement and occasional cleaning of enclosure surfaces. The system design successfully minimizes field service requirements while maintaining reliable operation.

Component reliability exceeds expectations with no significant failures observed during extended testing periods. The robust design approach and component selection prove effective in providing long-term operational stability.

4.5 System Strengths and Advantages

4.5.1 Technical Strengths

The system demonstrates several significant technical advantages compared to existing monitoring solutions. The tri-level fault detection capability provides unprecedented diagnostic granularity that enables efficient maintenance operations and reduces troubleshooting time.

Key Technical Advantages:

- **Granular Fault Diagnosis:** Distinguishes between multiple failure modes rather than simple on/off status
- **Non-invasive Installation:** Current transformer design requires no circuit modifications
- **Real-time Monitoring:** Continuous status updates with minimal latency
- **Scalable Architecture:** Cloud-based infrastructure supports unlimited device expansion
- **Battery Backup:** Maintains operation during commercial power outages
- **Weather-resistant Design:** IP65 enclosure suitable for harsh outdoor environments

The modular system architecture enables independent scaling of hardware, communication, and software components based on deployment requirements. This flexibility supports various implementation scenarios from small single-location deployments to large multi-regional networks.

4.5.2 Operational Benefits

The system provides substantial operational benefits that directly address the identified problems in outdoor advertising monitoring. Real-time fault detection significantly reduces downtime by enabling immediate response to equipment failures.

Operational Advantages:

- **Reduced Downtime:** Immediate fault notification enables rapid response
- **Lower Maintenance Costs:** Specific fault diagnosis reduces troubleshooting time

- **Improved Revenue Protection:** Faster failure detection minimizes advertising revenue loss
- **Enhanced Client Satisfaction:** Proactive monitoring demonstrates service reliability
- **Compliance Documentation:** Automated reporting supports regulatory requirements
- **Geographic Visualization:** Map-based interface enables efficient resource allocation

The automated reporting capabilities provide valuable business intelligence that supports operational decision-making and client relationship management. Historical data analysis enables identification of recurring problems and optimization of maintenance procedures.

4.5.3 Economic Impact

The system implementation provides measurable economic benefits through reduced operational costs and improved revenue protection. Faster fault detection and resolution directly translate to increased advertising uptime and revenue generation.

Economic Benefits:

- **Reduced Service Calls:** Specific fault diagnosis minimizes repeat visits
- **Optimized Maintenance Scheduling:** Predictive maintenance reduces emergency repairs
- **Improved Resource Allocation:** Geographic visualization enables efficient technician deployment
- **Enhanced Service Quality:** Real-time monitoring supports premium service offerings
- **Compliance Cost Reduction:** Automated documentation reduces administrative overhead

Return on investment calculations indicate payback periods of 12-18 months for typical advertising network deployments, demonstrating strong economic justification for system implementation.

4.6 Development Challenges and System Limitations

4.6.1 Hardware Development Challenges and Solutions

The hardware development process encountered several significant technical challenges that required innovative solutions and design modifications. The most critical challenge involved achieving reliable electrical isolation between high-voltage AC circuits and low-voltage microcontroller systems while maintaining measurement accuracy and signal integrity.

Optocoupler Integration Challenges: Initial designs experienced false triggering due to electrical noise, requiring extensive component testing and additional filtering circuitry. The

solution involved implementing three-stage voltage division using $75\text{k}\Omega$ resistors and bridge rectification with 4007 diodes to ensure clean signal transmission while maintaining electrical safety isolation.

Power Management Optimization: Managing the ESP32's varying current consumption patterns while providing stable regulation across temperature variations presented significant design challenges. The implemented solution utilized HI-Link switching power supplies with AMS1117 voltage regulation and TP4056 battery management, achieving stable operation with 6–7-hour backup capability.

PCB Design and Manufacturing: Transitioning from breadboard prototypes to manufactured PCBs required careful consideration of signal integrity, thermal management, and manufacturing constraints. Altium Designer enabled professional PCB layout with proper isolation between high-voltage and low-voltage sections while maintaining compact form factor.

4.6.2 Software Development Obstacles and Resolutions

Firmware development encountered significant challenges related to real-time performance requirements while maintaining reliable communication capabilities.

Multi-tasking Architecture Implementation: Initial single-threaded firmware experienced timing issues where communication activities interfered with sensor sampling accuracy. The solution involved migrating to FreeRTOS multi-tasking architecture with separate tasks for sensor acquisition, data processing, and communication, ensuring real-time responsiveness.

Communication Reliability Issues: MQTT communication reliability proved challenging in environments with unstable internet connectivity. Implemented solutions included message queuing, automatic reconnection logic with exponential backoff, and local data storage capabilities to handle temporary communication failures without data loss.

Memory Management Optimization: Long-running embedded applications required careful attention to prevent heap fragmentation and memory leaks. The solution employed static memory allocation strategies and careful pointer management to ensure stable operation over extended periods without memory-related failures.

Web Application Performance: Real-time data updates and map rendering efficiency required optimization for large datasets. Solutions included efficient WebSocket communication, client-side data caching, and progressive loading strategies that maintain responsive user interfaces even with hundreds of monitoring points.

4.6.3 System Integration Difficulties and Solutions

System integration presented complex challenges related to timing synchronization, data consistency, and error handling across multiple system components.

Database Schema Evolution: Managing database schema changes while maintaining data integrity required careful migration planning and version control strategies. The solution involved implementing backward-compatible schema updates with automated migration procedures.

Cross-platform Compatibility: Web application deployment revealed browser-specific compatibility issues requiring extensive testing and optimization. Solutions included polyfill implementations, responsive design testing, and performance optimization for various device types and network conditions.

Security Integration: Comprehensive authentication and authorization across multiple system components required balancing security with usability. The implemented solution utilized Firebase Authentication with role-based permissions and comprehensive audit logging throughout the system architecture.

4.6.4 Current System Limitations

Despite successful development challenge resolution, the final system exhibits certain limitations that constrain deployment and operational effectiveness.

Technical Constraints:

- **Wi-Fi Dependency:** System requires reliable wireless network connectivity, limiting deployment to areas with existing infrastructure
- **Battery Life Limitations:** Backup operation constrained to 6-7 hours during extended power outages
- **Content Monitoring Gap:** System verifies electrical operation but cannot detect display content issues
- **Environmental Sensitivity:** Battery performance degrades under extreme temperature conditions

Operational Limitations:

- **Installation Complexity:** Proper sensor placement and system configuration require technical expertise
- **Network Infrastructure Requirements:** Additional network equipment may be required for remote locations

- **Maintenance Access:** Enclosure design requires periodic access for battery replacement and system maintenance
- **User Training Dependencies:** Effective system utilization requires operator training on web interface and troubleshooting procedures

Cost Factors:

- **Initial Investment:** Hardware fabrication and professional installation costs may present barriers for smaller operators
- **Infrastructure Requirements:** Wi-Fi network establishment adds deployment complexity and expense
- **Ongoing Maintenance:** Battery replacement and system maintenance create recurring operational costs

4.7 Recommendations for Future Development

4.7.1 Communication Enhancement

Future system iterations should incorporate dual communication capabilities including cellular connectivity as backup to Wi-Fi networks. This enhancement would enable deployment in areas without reliable wireless infrastructure while providing communication redundancy.

Recommended Communication Improvements:

- **Cellular Backup:** NB-IoT or LTE-M module for areas without Wi-Fi
- **Satellite Communication:** For extremely remote locations
- **Mesh Networking:** Device-to-device communication for network resilience
- **Edge Computing:** Local data processing to reduce communication requirements

Implementation of hybrid communication systems would significantly expand deployment options while improving system reliability through communication redundancy.

4.7.2 Advanced Monitoring Capabilities

System functionality could be enhanced through additional monitoring capabilities that address current limitations and provide expanded diagnostic information.

Recommended Feature Additions:

- **Content Verification:** Camera or light sensor integration for display content monitoring
- **Environmental Monitoring:** Temperature and humidity sensing for preventive maintenance
- **Power Quality Analysis:** Voltage fluctuation and harmonic monitoring
- **Predictive Maintenance:** Machine learning algorithms for failure prediction
- **Remote Configuration:** Over-the-air parameter updates and system configuration

These enhancements would provide comprehensive monitoring capabilities while supporting predictive maintenance strategies that further reduce operational costs.

4.7.3 System Integration and Interoperability

Future development should focus on integration capabilities with existing business systems and industry-standard protocols to enhance operational efficiency.

Integration Recommendations:

- **ERP System Integration:** Automatic work order generation and maintenance scheduling
- **Mobile Application:** Dedicated mobile app for field technicians
- **API Development:** Third-party integration capabilities
- **Industry Protocol Support:** SNMP, Modbus, or other standard industrial protocols
- **Cloud Platform Integration:** Support for multiple cloud service providers

Enhanced integration capabilities would enable seamless incorporation into existing operational workflows while providing flexibility for diverse deployment requirements.

4.8 Project Contributions and Impact

4.8.1 Technical Contributions

This project makes several significant technical contributions to the field of outdoor advertising monitoring and IoT-based infrastructure management.

Primary Technical Contributions:

- **Novel Tri-Level Fault Classification:** First implementation of granular fault diagnosis specifically designed for advertising unit monitoring
- **Non-Invasive Monitoring Approach:** Innovative use of current transformers and optocoupler isolation for retrofit installation
- **Integrated IoT Architecture:** Comprehensive system design combining edge processing, cloud infrastructure, and web-based visualization
- **Scalable Monitoring Framework:** Architecture supporting unlimited expansion without infrastructure modifications

The technical approach demonstrates effective integration of embedded systems, IoT communication protocols, and cloud-based data management specifically tailored for advertising infrastructure monitoring.

4.8.2 Industrial Impact

The successful implementation provides a practical solution addressing critical needs in the outdoor advertising industry while demonstrating broader applicability to distributed infrastructure monitoring.

Industry Benefits:

- **Operational Efficiency:** Dramatic reduction in troubleshooting time and maintenance costs
- **Service Quality Improvement:** Enhanced reliability and client satisfaction
- **Revenue Protection:** Minimized advertising downtime through rapid fault detection
- **Competitive Advantage:** Advanced monitoring capabilities supporting premium service offerings

The system establishes new standards for advertising infrastructure monitoring while providing a foundation for similar applications in other distributed monitoring scenarios.

4.8.3 Academic and Research Value

The project contributes valuable knowledge to academic research in IoT system design, fault detection algorithms, and distributed monitoring architectures.

Academic Contributions:

- **Case Study Documentation:** Comprehensive implementation example for IoT monitoring systems
- **Performance Validation:** Quantitative results demonstrating system effectiveness
- **Design Methodology:** Systematic approach to distributed monitoring system development
- **Integration Strategies:** Practical examples of cloud infrastructure and embedded system integration

The detailed documentation and performance results provide valuable reference material for future research and development in similar application domains.

CHAPTER 5 : CONCLUSION

This chapter reflects on my six-month industrial training experience at Synergy Tech Consultancy (Pvt) Ltd., examining personal and professional development, project contributions, and alignment with training objectives. The training provided comprehensive exposure to modern IoT system development while addressing real-world industrial challenges in outdoor advertising infrastructure monitoring.

During my training at Synergy Tech Consultancy, I undertook the complete development of an Automated Functionality Detection System for Outdoor Advertising. My role encompassed end-to-end system development from initial problem identification through final deployment and validation. The project required integration of hardware design, embedded systems programming, cloud infrastructure implementation, and web application development. My primary responsibilities included designing tri-level fault detection circuits using PC817 optocouplers and current transformers, developing ESP32 firmware with FreeRTOS multitasking architecture, implementing MQTT communication protocols, and creating React.js web applications with Firebase authentication. I also handled PCB design using Altium Designer, 3D enclosure modeling with SolidWorks, and cloud infrastructure setup using InfluxDB and Telegraf for time-series data management.

The project achieved measurable success with 100% fault classification accuracy, 99.5% data transmission reliability, and 99.7% system uptime. The system successfully distinguishes between commercial power failures, DC converter malfunctions, and output switching circuit problems, providing maintenance personnel with specific diagnostic information that reduces troubleshooting time and operational costs.

Through this experience, I identified several personal strengths including strong analytical problem-solving abilities, adaptability to new technologies, and effective integration of multidisciplinary technical knowledge. My background in electronics provided a solid foundation for understanding complex circuit design and signal processing requirements. Areas requiring continued development include project management skills for larger-scale implementations, advanced debugging techniques for complex embedded systems, and deeper understanding of industrial manufacturing processes. Initial challenges with PCB layout optimization and power management circuit design highlighted the need for more practical experience in professional electronics development.

The training experience significantly enhanced my technical capabilities across multiple domains. Hardware development skills progressed from basic circuit understanding to professional PCB design and manufacturing. Software development abilities expanded from

simple programming to comprehensive full-stack web application development with modern frameworks and cloud integration. The exposure to industry-standard tools including Altium Designer, SolidWorks, STM32 microcontroller programming, MQTT protocols, React.js development, InfluxDB, Telegraf, and Grafana provided practical skills directly applicable to future engineering roles.

This training provided comprehensive preparation for roles in IoT system development, embedded systems engineering, and industrial automation. The interdisciplinary nature of the project developed systems thinking abilities essential for complex engineering challenges. Technical skills acquired during training directly align with current industry demands for IoT and embedded systems engineers. The experience of working with cutting-edge technologies while maintaining focus on reliable, practical solutions prepared me for the balance between innovation and implementation required in professional engineering roles.

The industrial training successfully achieved its primary objectives of providing practical engineering experience and professional skill development. The training environment at Synergy Tech Consultancy fostered independent learning while providing appropriate supervision and mentorship. The project's focus on addressing genuine industrial needs developed understanding of engineering's role in solving business problems and creating economic value, demonstrating successful integration of technical education with practical application.

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